

# Practical 6 - Competing risks 1

## Survival Analysis

London School of Hygiene & Tropical Medicine

Datasets required: `aaatrial.RData`

R packages required: `survival`, `cmprsk`

```
library(survival)
library(cmprsk)
```

### Introduction

In this practical we will use a dataset from a randomised trial of screening for abdominal aortic aneurysms (AAA), in which older men were randomised to be invited to be screened for AAA (invited group) or not (control group). The main outcome of interest was deaths relating to AAA (including deaths following AAA repair operations), but men in the study also died from other causes. There are 13,562 people in the data.

If an AAA was found at screening, a repair operation was carried out, essentially removing the risk of future death from AAA. However, the operation itself carries around a 5% risk of mortality. Undetected AAAs may rupture at any time, resulting in either an emergency repair operation (which carries around a 40% risk of mortality) or death. Participants were recruited to the trial between 1997 and 1999 and followed up until the end of 2007.

The variables used in this practical are described below.

| Variable  | Description                                                   |
|-----------|---------------------------------------------------------------|
| id        | Unique identifier for each participant                        |
| group     | Randomisation group: 0=Control, 1=Invited                     |
| dateran   | Date of randomisation                                         |
| aaadeath  | AAA-death indicator: 0=Censored or non-AAA-death, 1=AAA-death |
| allddeath | All-cause mortality: 0=Censored, 1=Death (any cause)          |

| Variable  | Description                              |
|-----------|------------------------------------------|
| deathtype | 0=Censored; 1=non-AAA death; 2=AAA-death |
| timeout   | Date of study exit                       |

This is already an R dataset, and the dates have been formatted for you as dates.

## Aims

The aims of this session are:

- Estimate cumulative incidence functions using non-parametric methods (the Aalen-Johansen estimator), and plot and interpret these.
- Compare cumulative incidence functions between groups.
- Understand why the Kaplan-Meier estimator is not in general suitable for estimating cumulative incidences when there are competing events.

### Question 1

- Load the data.
- How many people died from any cause, and how many had an AAA-death and a non-AAA-death?
- Calculate the follow-up time (since randomisation) in years using `dateran` and `timeout`, and call this variable `time`. This will be the time variable that we use in subsequent analyses.

### Question 2

We will begin by focusing on death from any cause using the `alldeath` variable.

- Plot the Kaplan-Meier survival curves for all-cause mortality.
- What is the risk of death from any cause up to 5 years?

### Question 3

We will now move on to considering AAA-deaths and non-AAA-deaths as competing events. We first explore how to use the `Surv()` function in R when there are competing risks.

Use the code below and investigate the output. What information is given for (i) people who have an AAA-death, (ii) people who have a non-AAA-death, (iii) people who are censored?

```
Surv(time=aaatrial$time,event=as.factor(aaatrial$deathtype))
```

### Question 4

- Estimate the cumulative incidences for AAA-deaths and non-AAA-deaths using

```
cinc <- survfit(Surv(time, as.factor(deathtype)) ~1, data=aaatrial)
```

- The resulting cumulative incidence curves can be plotted using

```
plot(cinc, lwd=2,col=c("black","grey"))  
legend("topleft",c("Non-AAA-death","AAA-death"),  
      lwd=2,col=c("black","grey"))
```

- What are the cumulative incidences for AAA-deaths and non-AAA-deaths up to 5 years?

### Question 5

- Returning to considering death from any cause (`alldeath`), obtain Kaplan-Meier plots by treatment group.
- What is the risk of death from any cause up to 5 years in the two groups?
- Use a log-rank test to test the null hypothesis that there is no difference between the survival curves in the two groups.

### Question 6

We now return to considering the competing events of AAA-death and non-AAA death.

- Estimate the cumulative incidence function for AAA-deaths and non-AAA-deaths in each treatment group using

```
cinc.group <- survfit(Surv(time, as.factor(deathtype)) ~ group, data=aaatrial)
```

- The resulting cumulative incidence curves can be plotted using

```
plot(cinc.group, lwd=2, lty=c(1,2,1,2), col=c("black", "black", "grey", "grey"))
legend("topleft", c("Control, Non-AAA-death", "Invited, Non-AAA-death",
                    "Control, AAA-death", "Invited, AAA-death"),
      lwd=2, lty=c(1,2,1,2), col=c("black", "black", "grey", "grey"))
```

- What are the cumulative incidences for AAA-deaths and non-AAA-deaths up to 5 years in the two groups?
- Perform Gray's test of the null hypothesis of no difference in the cumulative incidences for a given cause of death by group. This can be done using the `cuminc` function from the `cmprsk` package as follows:

```
cuminc(ftime=aaatrial$time, fstatus=aaatrial$deathtype, group=aaatrial$group)
```

EXTRA: Check out what is stored in `cinc.group$time`, `cinc.group$strata` and `cinc.group$pstate` and see if you can use this information to make a plot of the cumulative incidences of AAA-death only in the two groups.

## Question 7

- Use `Surv(time=aaatrial$time, event=aaatrial$aaaadeath)` and investigate the output. What information is given for (i) people who have an AAA-death, (ii) people who have a non-AAA-death, (iii) people who are censored? Compare this with what you obtained in Question 3 where you looked at `Surv(time=aaatrial$time, event=as.factor(aaatrial$deathtype))`.
- Use the Kaplan-Meier estimator to estimate the cumulative incidence of AAA-death using

```
cinc.km <- survfit(Surv(time, aaadeath) ~ 1, data=aaatrial)
```

- Plot the cumulative incidence function based on `cinc.km` and compare this with what you obtained in `cinc` (i.e. based on the Aalen-Johansen estimator). Compare the estimates and comment on why the results are different.